# OTEPIC

## CARBON FARMING


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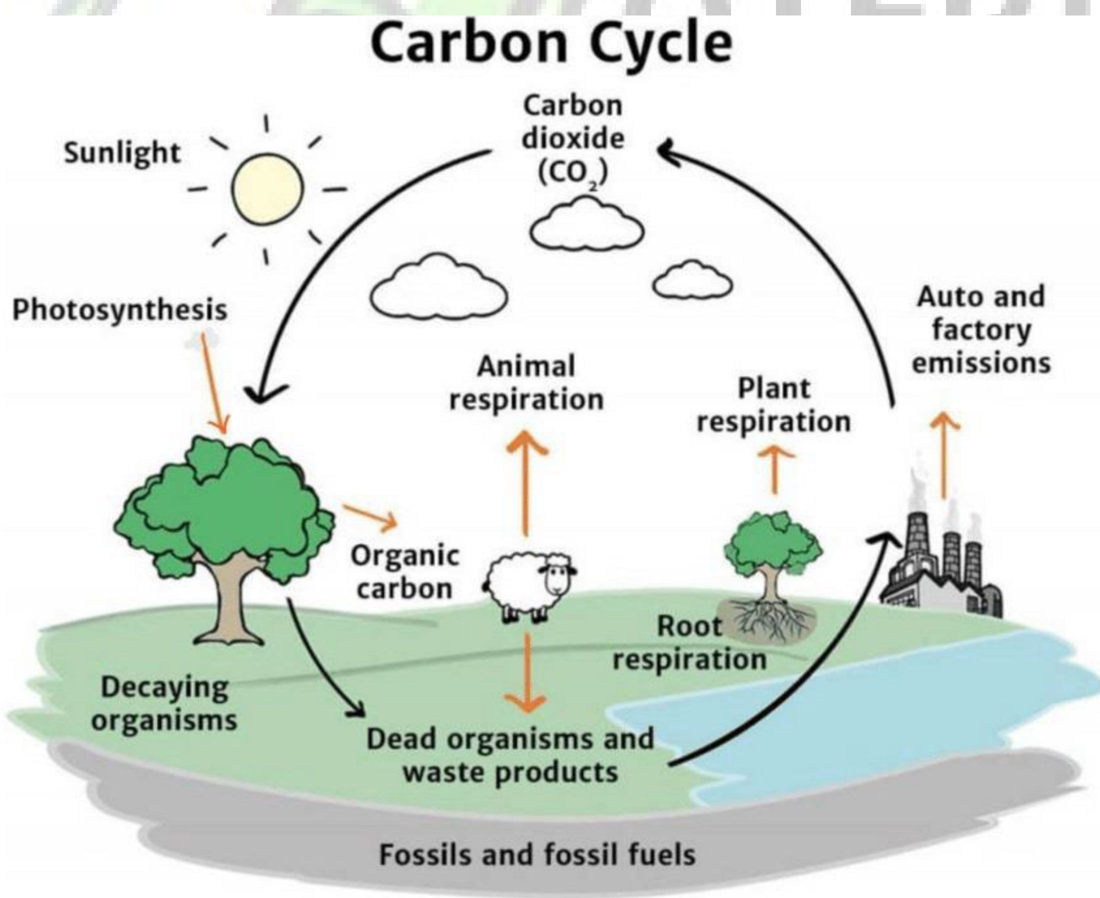
**TRAINING MANUAL HANDBOOK**

## CARBON FARMING

refers to agricultural practices that increase the amount of carbon sequestered in soil and vegetation, contributing to climate change mitigation. Here are the key advantages of carbon farming:

Its a farm approach that is aimed at optimizing carbon capture on working landscapes by implementing practices known to improve the rate at which $CO_2$ is removed from the atmosphere and stored in plant material or soil organic matter. It is a framework for engaging with agroecosystem processes that drive system change. It recognizes that it is solar energy that drives farm ecosystem dynamics and that carbon is the carrier of that energy within the farm system. Carbon farming uses the term" regenerative agriculture" when that term is explicitly rooted in an understanding of the underlying system dynamics and positive feedback processes that actually make a "regenerative" upward spiral of soil fertility and farm productivity possible, as depicted in the figure below.



Carbon farming can also point to set of farm practices that when applied reduces greenhouse gas emissions and store carbon in the soil, crop roots, wood, and leaves. its goal is to reduce carbon in the atmosphere.

Achievements of Carbon Farming

- **Improve soil health:** Carbon farming can improve soil fertility and resilience.
- **Promote biodiversity:** Carbon farming can promote biodiversity and increase the resilience of natural systems.
- **Mitigate climate change:** Carbon farming can help mitigate climate change by reducing greenhouse gas emissions.
- **Create income:** Carbon farming can create income for farmers through the sale of carbon credits.

**Here are the key advantages of carbon farming:**

1. **Climate Change Mitigation:** Carbon Sequestration: Carbon farming practices, such as agroforestry, cover cropping, and no-till farming, capture carbon dioxide from the atmosphere and store it in the soil and plants. This helps reduce the overall concentration of greenhouse gases in the atmosphere, contributing to the fight against climate change.
2. **Reduction of Greenhouse Gas Emissions**: By implementing practices that enhance soil carbon storage, farmers can help offset emissions from other sectors, making agriculture a key player in mitigating climate change.
3. **Soil Health Improvement:** Enhanced Soil Fertility: Practices like adding organic matter, composting, and rotating crops improve soil structure, increase nutrient availability, and enhance biological activity, leading to healthier, more productive soils.
4. **Increased Organic Matter:** Carbon farming boosts soil organic matter, which improves water retention, aeration, and microbial diversity, making soil more resilient to droughts and floods.
5. **Increased Agricultural Productivity**: Improved Crop Yields: By enhancing soil health through carbon farming techniques, crops often experience higher yields due to better nutrient cycling, moisture retention, and reduced soil erosion. Healthy soils support robust plant growth, leading to more productive and sustainable farming systems.

6. Stronger Plants: Healthy soils enriched with organic carbon leads to stronger, more resilient plants that are less susceptible to pests, diseases, and extreme weather conditions.
7. **Water Retention and Drought Resilience:** Improved Water Holding Capacity: Carbon-rich soils have better water retention capacity, reducing the need for irrigation and improving resilience during dry spells or droughts. This is particularly important in areas facing water scarcity or unpredictable weather patterns.
8. **Reduced Irrigation Costs:** With better water retention in the soil, farmers can reduce their reliance on irrigation systems, cutting down water use and saving costs associated with irrigation.
9. **Reduced Soil Erosion:** Improved Soil Structure: By increasing the organic carbon content, soil becomes more stable and less prone to erosion. This is crucial for preserving topsoil, which is vital for long-term agricultural productivity.
10. **Cover Cropping and Reduced Tillage**: Techniques like cover cropping and reduced tillage, key elements of carbon farming, reduce soil disturbance, preventing erosion caused by wind and water.
11. **Economic Benefits and New Revenue Streams:** Carbon Credit Market: Carbon farming can enable farmers to participate in carbon credit programs, where they can earn payments for carbon sequestration. By selling carbon credits, farmers can access a new revenue stream, helping to make farming more profitable and sustainable.
12. **Cost Savings:** Practices like reduced tillage, agroforestry, and cover cropping can reduce the need for synthetic fertilizers, herbicides, and pesticides, lowering input costs for farmers.
13. **Biodiversity Enhancement/Diverse Farming Systems:** Carbon farming often involves practices like agroforestry, intercropping, and incorporating wildlife corridors, which enhance biodiversity by creating diverse habitats for plants, animals, and insects.
14. **Increased Pollinator Habitats**: Practices such as planting cover crops or maintaining hedgerows can provide habitats for pollinators like bees, which are essential for crop pollination and overall ecosystem health.
15. **Sustainable Land Management/Restoration of Degraded Land:** Carbon farming practices can help rehabilitate degraded soils and lands that have been subjected to overuse or poor management. By focusing on carbon sequestration, these practices help restore productivity, making land use more sustainable in the long term.

16. **Agroecological Practices:** Carbon farming promotes agroecological practices that work in harmony with nature, reducing dependency on chemical inputs and enhancing the long-term sustainability of farming systems.
17. **Climate Resilience:** Enhanced Resilience to Extreme Weather: By improving soil structure, water retention, and biodiversity, carbon farming helps farms better withstand extreme weather events such as heavy rains, floods, and droughts, making them more resilient

ways of practicing carbon farming:

- **Reduce tillage**: Frequent, heavy tilling releases carbon dioxide from the soil and breaks up soil structure, which can lead to erosion. Regenerative tillage, which involves minimal or no tilling, helps preserve soil quality and carbon.
- **Use cover crops**: Cover crops are one way to sequester carbon in the soil.


**Improve residue management**: Managing crop residue is a carbon farming technique that can be tailored to a farm's specific needs.

- **Incorporate organic matter**: Maintaining ground cover and incorporating organic matter can help with carbon farming. Eg, Compost


- **Plant trees**: Planting trees is one carbon farming activity.
- **Change grazing patterns**: Changing grazing patterns and herd management can be part of carbon farming.
- **Change manure management**: Changing manure management can be part of carbon farming.
- **Change soil management**: Changing soil management can be part of carbon farming.
- **Change fertilizer use**: Changing how fertilizer is used can be part of carbon farming.